

# Graphs and signed graphs with integer reciprocal eigenvalue properties

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Joint work with Francesco Belardo (University of Naples Federico II, Italy)  
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*Dedicated to Professor Vilmar Trevisan.*

# The properties

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Setting  $m = 1$  recovers the classical (R), (SR), (−R) and (−SR).

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For signed graphs it is strictly weaker: there are non-bipartite signed graphs with symmetric spectrum.

## Products, coronae, hypercubes

# Kronecker products

Following Germina, Hameed and Zaslavsky, the Kronecker product  $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$  has vertex set  $V_{\Gamma_1} \times V_{\Gamma_2}$ , with  $(u_i, v_j) \sim (u_r, v_s)$  when  $u_i \sim u_r$  and  $v_j \sim v_s$ , taking the signature  $\sigma_1(u_i u_r) \sigma_2(v_j v_s)$ .

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### Corollary

*With an abuse of notation...  $m_1$ -(R)  $\otimes$   $m_2$ -(−R) gives  $m_1 m_2$ -(−R), and  $m_1$ -(−R)  $\otimes$   $m_2$ -(−R) gives  $m_1 m_2$ -(R).*

The strong versions survive too, pairings in each factor induce a multiplicity-preserving bijection, so repeated products are fine.

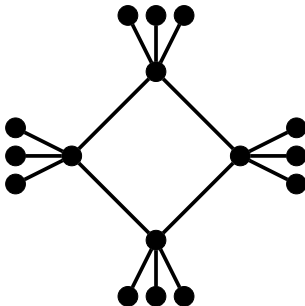
## Signed simple $m$ -coronae

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For example, here we have the simple 3-corona of  $C_4$ :



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Compare Barik–Pati–Verma, who obtain  $m$ -(−R) in general and both strong properties for bipartite roots. So this strengthens the conclusion and weakens the hypothesis simultaneously.

# Signed simple $m$ -coronae: the argument (1/2)

With a suitable ordering of the vertices,

$$A_{((\Gamma, \sigma) \circ \cup_m (K_1, +))} = \begin{pmatrix} A_{(\Gamma, \sigma)} & I_n \otimes \mathbf{1}_m^T \\ I_n \otimes \mathbf{1}_m & O_{mn} \end{pmatrix},$$

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So non-zero  $\lambda$  is an eigenvalue of the simple  $m$ -corona precisely when  $\lambda - m/\lambda$  is an eigenvalue of the root graph.

## Signed simple $m$ -coronae: the argument (2/2)

Solving  $\lambda^2 - \mu\lambda - m = 0$  for each root eigenvalue  $\mu$  gives

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The identity divides by  $\lambda$ , so the nullity is found separately. Each root vertex contributes  $m - 1$  pendant vectors summing to zero, giving a nullity of  $n(m - 1)$  whatever the root.

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So,

$$\text{spec}(Q_1, +) = \{1, -1\},$$

$$\text{spec}(Q_2, +) = \{2, 0^{(2)}, -2\},$$

$$\text{spec}(Q_3, +) = \{3, 1^{(3)}, -1^{(3)}, -3\},$$

$$\text{spec}(Q_4, +) = \{4, 2^{(4)}, 0^{(6)}, -2^{(4)}, -4\},$$

and  $Q_1$  satisfies 1-(SR),  $Q_2$  satisfies 4-(SR),  $Q_3$  satisfies 3-(R), and  $Q_4$  satisfies 8-(R), each with its anti-reciprocal counterpart.

# Hypercubes proposition

## Proposition

*Let  $n \geq 1$ . There is a positive integer  $m$  with  $Q_n$  satisfying  $m$ -(R) if and only if  $n \leq 4$ . The same holds for  $m$ -(-R).*

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But  $\text{lcm}(n, n-2, n-4)$  grows cubically and outruns  $n^2$ .

Only  $n \in \{5, 6, 8, 12\}$  need checking by hand, and each fails. □

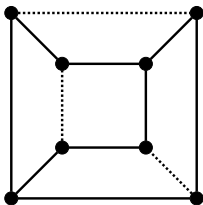
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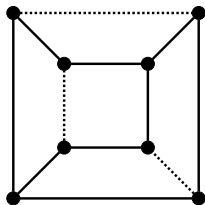
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Such a signed hypercube has two distinct eigenvalues,  $\sqrt{n}$  and  $-\sqrt{n}$ , each with multiplicity  $2^{n-1}$ . This is equivalent to  $A_{(Q_n, \sigma_n)}^2 = nI$ .

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Introducing a (suitable) signature therefore gives properties that the unsigned hypercube cannot have for any  $n \geq 5$ .

## Graphs satisfying $m$ -(R) but not $m$ -(SR)

## Separating $m$ -(R) from $m$ -(SR)

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Write

$$\Omega_1^m = \{\Gamma_{m,n} : m, n \in \mathbb{N}, m \geq 1, n \geq 2\}.$$

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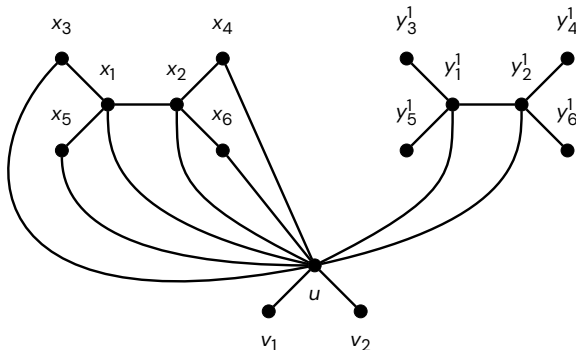
*Set  $R = R_1 \circ \cup_m K_1$  with  $R_1 = K_1$ ,  $V_{R_1} = \{u\}$ . Let  $\Gamma_{m,n}$  with  $m \geq 1$  and  $n \geq 2$ , be obtained by taking  $n$  copies of  $K_2 \circ \cup_m K_1$  and one copy of  $R$ , then joining  $u$  to every vertex of the first copy of  $K_2 \circ \cup_m K_1$  and to the vertices of degree  $m + 1$  in the remaining  $n - 1$  copies.*

*Write*

$$\Omega_1^m = \{\Gamma_{m,n} : m, n \in \mathbb{N}, m \geq 1, n \geq 2\}.$$

Setting  $m = 1$  recovers the Panda–Pati (2016) construction exactly.

# A small example



The graph  $\Gamma_{2,2} \in \Omega_1^2$ , which satisfies 2-(R) but not 2-(SR).

# The characteristic polynomial

## Theorem

Let  $\Gamma_{m,n} \in \Omega_1^m$ . Then  $\Gamma_{m,n}$  satisfies  $m$ -(R) but not  $m$ -(SR), and

$$\begin{aligned}
 P(\Gamma_{m,n}; \lambda) = & \underbrace{(\lambda^{2m+2} - (2m+1)\lambda^{2m} + m^2\lambda^{2m-2})^{n-1}}_{\text{from the } n-1 \text{ outlying copies}} \\
 & \times \left\{ \lambda^{3m+3} - (2n+5m+1)\lambda^{3m+1} - (2n+4m)\lambda^{3m} \right. \\
 & \quad \left. + m(2n+5m-1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3} \right\}.
 \end{aligned}$$

# The characteristic polynomial

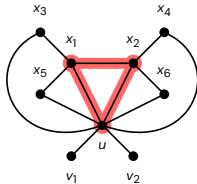
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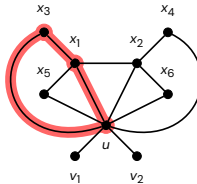
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The computation runs on two classical formulas of Schwenk: one stripping away pendant vertices, and one expanding about  $u$  over all cycles through it.

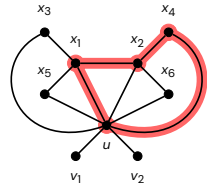
Every cycle of  $\Gamma_{m,n} - \{v_1, \dots, v_m\}$  passes through  $u$ , and these cycles fall into six types, counted 1,  $2m$ ,  $2m$ ,  $m^2$ ,  $m(m-1)$ , and  $n-1$  times respectively.



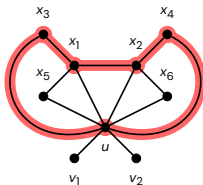
Type (i)



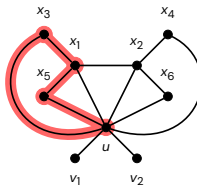
Type (ii)



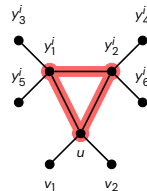
Type (iii)



Type (iv)



Type (v)



Type (vi)

# Why $m$ -(R) holds and $m$ -(SR) fails

Write the two bracketed factors of  $P(\Gamma_{m,n}; \lambda)$  as  $F_1(\lambda)$  and  $F_2(\lambda)$ .  
Their roots are

$$\lambda_{(1,\pm)} = \frac{1}{2}(1 \pm \sqrt{4m+1}), \quad \lambda_{(2,\pm)} = \frac{1}{2}(-1 \pm \sqrt{4m+1}),$$

from  $F_1$ , together with  $\lambda_{(3,\pm)} = \lambda_{(2,\pm)}$  and two further pairs  
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The failure is one of multiplicity. Since  $\lambda_{(2,\pm)} = \lambda_{(3,\pm)}$ , these are roots of *both*  $F_1$  and  $F_2$ , while  $\lambda_{(1,\pm)}$  is a root of  $F_1$  only. Hence

$$\text{mult}(\lambda_{(2,\pm)}) = n \quad \text{but} \quad \text{mult}(\lambda_{(1,\pm)}) = n - 1,$$

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# A signed variant

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As  $A_{(\Gamma,-)} = -A_\Gamma$ , the spectrum is negated and the conclusion is immediate:  $(\Gamma_{m,n}, -)$  also satisfies  $m$ -(R) but not  $m$ -(SR).

## Non-corona graphs satisfying $m$ -(SR) and $m$ -(-SR)

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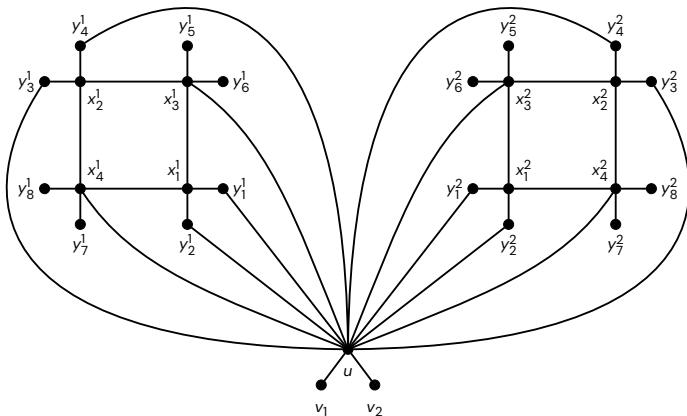
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Setting  $m = 1$  recovers the class  $\mathcal{NC}_1^1$  of Rakshith, Das and Manjunatha (2022).

# The family $\Omega_2^m$



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# Theorem and sketch proof

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Write  $\begin{pmatrix} O_k & C \\ C^T & O_k \end{pmatrix}$  for the adjacency matrix of  $C_{2k}$ , and let  $\lambda_i$  be the singular values of  $C$ . Eigenvectors built from the singular vector pairs of  $C$  contribute

$$c_i^\pm = \frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2}, \quad d_i^\pm = \frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2},$$

with  $c_i^+ c_i^- = -m$  and  $c_i^+ d_i^+ = m$ , giving both pairings at once.

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An equitable partition into six cells accounts for the remaining eigenvalues, which satisfy

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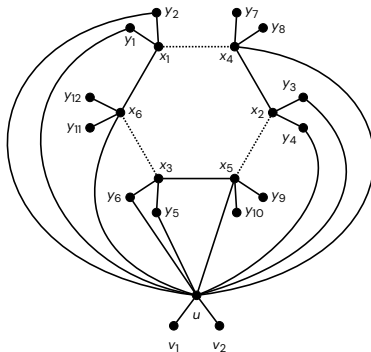
The constant term is  $m^2$ , so these pair correctly too.

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Let  $(C_{2k}, \tau)$  be the signed cycle in which  $\tau$  is  $-1$  on the perfect matching  $\{\{x_i, x_{k+i}\}\}$  and  $+1$  on the complementary matching. Take all remaining edges positive.

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$(\Gamma_{3,2,1}, \sigma) \in (\Omega_2^2, \Sigma)$ : non-corona, unbalanced, satisfying 2-(SR) and 2-(-SR).

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The constant term is still  $m^2$  and the pairing survives untouched.

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- $\Omega_1^m$  and its signed variant separate  $m$ -(R) from  $m$ -(SR), by a multiplicity mismatch of exactly one.
- $\Omega_2^m$  gives non-corona graphs with both strong properties, and admits an unbalanced signed variant.

*Thank you!*

*Grazie!*

*Obrigado!*